The magnetic susceptibility of various compounds/elements at room temperature and the Curie temperature of solid Gadolinium

Ben *Morrison*

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Abstract

Guoy's method was used to calculate the mass susceptibility, χ_{mass}, of 1 liquid and 11 solid samples at room temperature. To do this, the magnetic field of a permanent magnet was mapped out in space. The samples measured were: Copper [Cu], Aluminium [Al], Titanium [Ti], Bismuth [Bi], Iron Alumn [$NH_4Fe(SO_4)_2\,12\,H_2O$], Blue Vitriol [$CuSO_4\,5\,H_2O$], Water [H_2O], Mohr's Salt [$(NH_4)_2Fe(SO_4)_2\,6\,H_2O$], Neodymium Chloride Hexahydrate [$NdCl_3\,6\,H_2O$], Manganese Oxide [MnO], Manganese Chloride Tetrahydrate [$MnCl_2\,4\,H_2O$], and Copper Acetate [$Cu(OAc)_2\,H_2O$]. The Curie temperature for Gadolinium [Gd] was determined via a plot of χ_{mass} vs T where χ_{mass} was obtained by performing Guoy's method at a range of temperature values achievable with tap water ($\approx 40°C$) and a minifridge ($\approx 6°$C).

1 Results

1.1 Magnetic field of the permanent magnet

The magnetic field strength was recorded as a function of distance and plotted in Fig. 1. The sample was axially aligned to the centre of the top face (x, y plane) of the permanent magnet so that the magnetic field need only be measured in the axis perpendicular to that face (z axis).

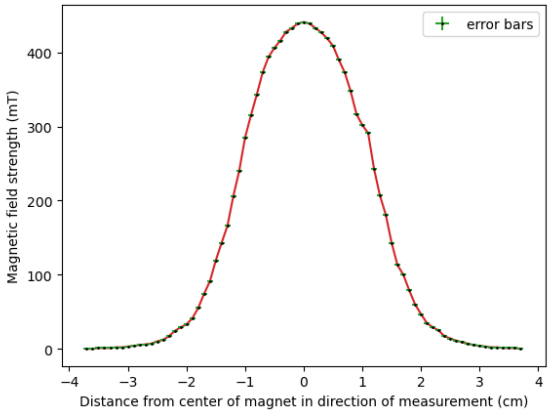


Figure 1: Magnetic field of the permanent magnet plotted as a function of temperature. The point $z = 0$ corresponds to the centre of the magnet in all 3 dimensions (x, y, z). Error bars are included in both axes.

1.2 Mass Susceptibilities

The results for χ_{mass} are displayed in table 1, where they are compared to the literature values if possible, and as a histogram in Fig. 2.

1.3 Curie Temperature of Solid Gadolinium

The mass susceptibility of solid Gadolinium was plotted as a function of temperature between $10 \pm 0.01°C$ and $37.8 \pm 0.01°C$, Fig. 3.9 The value of T_c is read off this graph as the point where the graph begins to decrease rapidly, which is $\approx T_c = 18.5 \pm 0.5°C$.

2 Discussion

2.1 Permanent Magnet

2.1.1 Uncertainties

The B field mapped out in Fig. 1 was measured using a hall probe and a ruler. The corresponding uncertainties were $\sigma_{ruler} = \pm 0.05cm$ and $\sigma_{hp} = \pm 1mT$ - both values coming from the reading uncertainty of the instrument. When the height of a sample did not directly correspond to

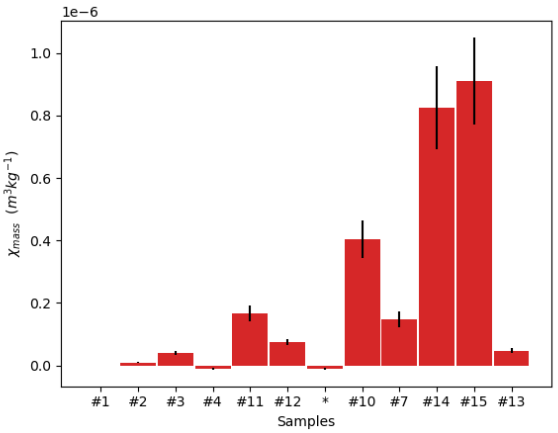


Figure 2: Samples are numbered in the same order as in the Techspin manual appendix B [1]: #1 - Copper; #2 - Aluminium; #3 - Titanium; #4 - Bismuth; #5 - Iron Alum; #7 - Neodymium Chloride Hexahydrate; #10 - Mohr's Salt; #11 - Iron Alum; #12 - Iron Alum; #13 - Copper Acetate; #14 - Manganese Oxide; #15 - Manganese Chloride Tetrahydrate; * - Water.

one of the ruler's measurements, the value for B was read off the graph. This introduces a separate uncertainty for these specific values. To quantify this uncertainty it was assumed that the regions between two points is linear and thus can be modelled as:

$$B \approx Mz + c \tag{1}$$

where M is approximated as:

$$M \approx \frac{\Delta B}{\Delta z} = \frac{B_2 - B_1}{z_2 - z_1} \tag{2}$$

with B_2, B_1, z_2 and z_1 being the points immediately surrounding the point being read off the graph. This means that the error for the interpolated values is:

$$\sigma_{interp} \approx M\sigma_{ruler} \tag{3}$$

When the measurements for the magnetic field strength were originally taken the hall probe was not set to it's highest precision mode (in err). This was only realized when there was insufficient time to reproduce the measurements. Had the higher precision mode being used the uncertainty on the B field measurements could be reduced from $\sigma_{hp} = \pm 1mT$ to $\sigma_{hp} = \pm 0.1mT$. If the experiment is repeated this should be considered.

2.1.2 Interpretation of results

Qualitatively, the graph for the B field seems appropriate as it has a maximum value at z=0 and decreases $\sim \frac{1}{z^2}$ as the distance from the centre increases. Quantitively, the error bars for the points are so small they are almost illegible on the graph (however, they have still been included). In the absence of any apparent systematic uncertainty the data seems to accurately represent the B field in the axis of measurement between the two permanent magnets.

2.2 Mass Susceptibilities

Mass susceptibilities were calculated using:

$$\chi_{mass,1} = \frac{\Delta m}{m} \frac{2\mu_0 gh}{(|\underline{B}(h)|^2 - |\underline{B}(0)|^2)} \tag{4}$$

and

$$\chi_{mass,2} = \frac{\Delta m}{m} \frac{2\mu_0 gh}{(|\underline{B}(0)|^2 - |\underline{B}(h)|^2)} \tag{5}$$

for the respective measurement positions 1 and 2 (note: the susceptibilities for all tested samples in position 3 and 4 were $\chi_{mass} = 0 \pm \sigma_{\chi,i}$ where $\sigma_{\chi,i}$ is the assoicated uncertainty for each sample from table 1).

2.2.1 Uncertainties

The uncertainties on χ_{mass} were calculated using the Monte Carlo error propagation technique. For each variable α a Gaussian distribution centred on the mean value of α, with width equal to the experimental error on α, is modelled. A set of random values, $\{\alpha_i\}$, are then selected from the distribution ($\approx 10^6$ values). The set of values for each variable are then passed into equations 4 and 5 which yields two sets of mass susceptibility values, $\{\chi_{mass,1,i}\}$ and $\{\chi_{mass,2,i}\}$. These sets are then averaged and the standard deviation is taken - this is the final quoted uncertainty for each sample. The source code for these calculations will be included in the submission, (*guoy_values.py*). When searching for an experimental error on g it was decided to set the value to 0 since the exact value of g will vary based on the altitude of the location where experiments were conducted. If the value of g was determined experimentally at the same location prior to the susceptibility measurements a value for σ_g could have been determined. The only alternative to setting $\sigma_g = 0$ is to hazard a guess - both options are inaccurate. However, since the value of g is widely accepted to be

9.80665, [2], for the purpose of this experiment g has been rounded to $g = 9.81 \pm 0 ms^{-2}$. The error on μ_0 is so comparatively low that it is also considered to be 0, [3].

2.2.2 Interpretation of results

From table 1 it can be seen that the value of χ_{mass} for Copper, Aluminium, Titanium, Iron Alum, Water, Mohr's Salt, Manganese Oxide, Manganese Chloride Tetrahydrate and Copper acetate are all in close agreement with the literature values (within 2σ). A literature value for Neodymium Chloride Hexahydrate could not be found and thus the accuracy of this result is unknown. Manganese Oxide and Managanese Chloride Tetrahydrate have the highest susceptibilities of all the samples and thus a very high uncertainty is attributed to these values. The high susceptibility meant that the samples consistently stuck to the sides of the magnets when attempting to measure positions 1, 2 and 4. When this happened the apparatus had to be realigned (by eye) and then the measurement repeated. To reduce this uncertainty two changes could be made to the experimental apparatus:

1. Increase the distance between the two permanent magnets
2. Realign the apparatus with a ruler

These changes would decrease the likelihood of the sample getting stuck to the permanent magnet and, if the sample were to get stuck, ensure that all measurements for the sample are axially aligned. These changes would come with payoffs - separating the magnets will decrease the strength of the magnetic field along the axis of measurement and thus increase the relative uncertainty on the measurement of said field. If the apparatus is realigned with a ruler this will take significantly longer than by eye, reducing the number of samples that can be measured within the prescribed laboratory time. The value for Iron Alum is of the same order of magnitude as the literature value but out by a factor of ≈ 2, which is $\approx 9\sigma_\chi$. This is likely due to human error during laboratory time (wrong numbers recorded, wrong sample, scale not zeroed, etc). An incorrect input into a spreadsheet is highly plausible because the ability of the scale to directly print numbers into said spreadsheet was discovered after all the

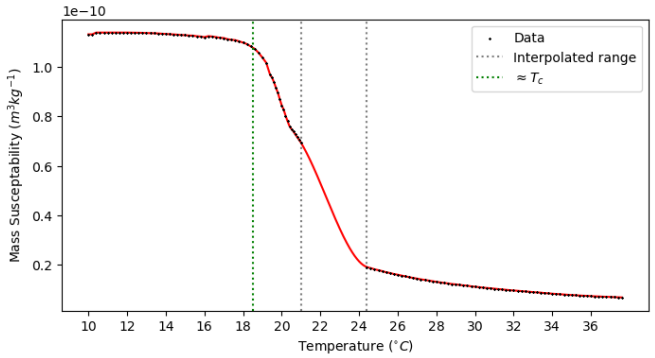


Figure 3: Plot of mass susceptibility, χ_{mass}, and temperature in the range $10°C < T < 37.8°C$ with no data points in the range $21°C < T < 24.4°C$ because the sample was cooling/heating very slowly as it approached these temperatures. An interpolation for this region was performed on python (*curie_weiss.py*).

χ_{mass} measurements were made. To improve this experimental procedure in the future this should be taken into account. The value for the room temperature volume susceptibility of Iron Alum is well documented, [4], and great care has been taken to ensure the units are being converted correctly, thus it can be said with confidence that the result obtained in this experiment does not accurately represent the mass susceptibility of Iron Alum.

2.3 Curie Temperature of Gadolinium

2.3.1 Uncertainties

The curie temperature was read off of the plot shown in Fig. 3 as the point where the graph begins to steeply decline. The temperature values were measured with a digital thermometer with uncertainty $\sigma_T = \pm 0.1°C$. The uncertainty in χ_{mass} was calculated, like in section 2.2.1, using the Monte Carlo method (*curie_weiss.py*). The uncertainty becomes much lower in the region $19.1°C < T < 21.0°C$ as the measurement interval for temperature was reduced to $0.1°C$. This was done because the Δm values began to change much faster than outwith the region. During this part of the experiment the values for temperature measured were not that of the Gadolinium sample but the temperature of the water that it was immersed in. Due to the low amount of water in the sample it is likely that these temperatures vary by some amount (due to having different heat capacities, more on this in section 2.3.2). By inspecting

the plot it can be seen that the point where the graph steeply declines is not easily determined. A good approximation is somewhere in the range $18°C < T < 19°C$ - thus the value for T_c is quoted as $T_c = 18.5 \pm 0.5°C$ (the error on reading the plot is much larger than from the experimental errors).

2.3.2 Interpretation of results

The literature value for the Curie temperature of Gadolinium is $239.2K$ $(20.05°C)$ [5]. The value from this experiment agrees within 4σ. This is likely due to the fact that water and gadolinium have different specific heat capacities. As a result they will heat up and cool down at different rates. Since water has a higher specific heat capacity than Gadolinium, [6, 7], the gadolinium would have been slightly cooler than the water within the temperature range $(T > 24.4°C)$ and slightly warmer than the water in the temperature range $(T < 21.0°C)$. The value of T_C is in the latter region and thus, assuming the temperature discrepancy is $\sim 1°C$, more likely within 2σ of the literature value. One way to improve on this procedure is to insert an axial hole into the gadolinium sample so that the thermometer can be in direct thermal contact with the gadolinium. This has the consequence that the gadolinium sample will have to be left in the fridge for a longer period to cool down and take much longer to heat back up to room temperature. To make this process suitable for the 4 hour lab sessions available, data collection could be automated using a computer (perhaps left overnight or initiated at the beginning of the lab session) so that this experiment does not consume a large portion of the laboratory time.

2.4 Consequences

All of the substances tested are widely used in semiconductors[6], water treatment [8], fertilizers [9], nuclear reactors [6] and many other applications. Naturally these materials will be in environments with magnetic fields and so the ability to predict their behaviour precisely is crucial for the health and safety of people and animals that come into contact with them regularly. Further work could be conducted in extending the experiment to plot mass susceptibility as a function of temperature for all samples. This would benefit the sectors which operate outwith normal room temperature conditions (outer space, nuclear power plants, etc.).

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Sample	$\chi_{\text{mass}}(\text{m}^3\text{kg}^{-1} \cdot 10^{-7})$	
	This experiment	Literature
Copper [10]	−0.008(2)	-0.012
Aluminium [10]	0.093(18)	0.078
Titanium [10]	0.400(62)	0.401
Bismuth [10]	−0.113(25)	-0.17
Iron Alum [11], [4]	1.66(25)	3.860
Blue Vitriol [8]	0.736(10)	0.669
Water [12]	-0.103(25)	-0.091
Mohr's Salt [12]	4.04(61)	4.02
Neodymium Chloride Hexahydrate	1.47(26)	N/A
Manganese Oxide [13]	8.3(13)	8.621
Manganese Chloride Tetrahydrate [14]	9.1(14)	9.270
Copper Acetate [15], [16]	0.474(73)	0.508

Table 1: Mass magnetic susceptibility for all 12 samples from literature and this experiment. Sources for the literature values are cited with sample names. All literature values are from room temperature measurements.